

Gravitational waves from a classically scale-invariant dark $U(1)_D$ first-order phase transition and their confrontation with pulsar-timing-array data

DarkAgents

2026-06-09

Abstract

We report an end-to-end **fopt-pt** pipeline analysis of a classically scale-invariant (conformal) dark $U(1)_D$ extension of the Standard Model (SM), in which a complex scalar Φ and a dark gauge boson A' undergo Coleman–Weinberg (CW) radiative symmetry breaking. We compute the stochastic gravitational-wave background (SGWB) from the resulting strongly supercooled first-order phase transition (FOPT), confront the spectrum with the NANOGrav 15-year (NG15) dataset through a PTArcade Bayesian inference campaign, and audit the laboratory, astrophysical and cosmological constraints on the preferred region, together with the dominant modelling assumptions. The campaign favours a dark vacuum expectation value (vev) $v \simeq 1.09$ GeV and gauge coupling $g_D \simeq 0.528$ (MAP), corresponding to a GeV-scale dark photon. No constraint currently excludes the 1σ region, but the preferred region is *provisional*: it rests on a no-supermassive-black-hole-binary (SMBHB) foreground fit, an order-of-magnitude gauge-dependence theory uncertainty on $h^2\Omega_{\text{GW}}$, an unresolved $\xi = 1$ versus $\epsilon \simeq 0$ internal tension, and four high-priority constraint calculations that remain to be performed.

1 Prompt

The verbatim user request was: “*can you study the model in `input/u1conformal/prompt.md`?*” The referenced specification describes a classically scale-invariant/conformal dark $U(1)_D$ extension of the SM (complex scalar Φ plus dark gauge boson A' , CW radiative breaking, free parameters v and g_D), and requests: (i) analysis of the GW spectrum from a FOPT and comparison to PTA data; (ii) preparation of a PTArcade inference campaign and extraction of the parameter-space region probeable by PTArcade; (iii) a check of all collider/astrophysical/cosmological constraints; and (iv) identification of assumptions, uncertainties and caveats relevant for interpreting the results. The orchestrator ran the full **fopt-pt** pipeline, skipped the **proposer-agent** (the model is self-contained), and proceeded past the critic’s non-blocking clarifications with standard conformal-CW defaults.

2 Model description (critic validation)

The dark-sector Lagrangian is

$$\mathcal{L}_D = |D_\mu \Phi|^2 - \lambda_\Phi |\Phi|^4 - \frac{1}{4} F'_{\mu\nu} F'^{\mu\nu}, \quad D_\mu = \partial_\mu - ig_D A'_\mu, \quad (1)$$

with $\Phi(x) = (\chi + \rho(x) + iG(x))/\sqrt{2}$, where χ is the classical background field, ρ the radial mode (scalon) and G the Goldstone mode eaten by A' . The tree potential is $V_{\text{tree}}(\chi) = \lambda_\Phi \chi^4/4$ with no tree-level mass term, so the theory is classically scale invariant; scale invariance is broken

radiatively (CW dimensional transmutation). The field-dependent gauge-boson mass is $m_{A'}^2(\chi) = g_D^2 \chi^2$, and the radiatively generated radial mass is

$$m_\rho^2 = \beta_\lambda v^2, \quad \beta_\lambda = \frac{1}{8\pi^2} \left(\sum_b n_b g_b^4 - \sum_f n_f g_f^4 \right) = \frac{3g_D^4}{8\pi^2}, \quad (2)$$

with the only contribution from the dark gauge boson ($n_b = 3$, $g_b = g_D$). The optional Higgs portal $\lambda_{H\Phi}|H|^2|\Phi|^2$ and kinetic mixing $(\epsilon/2)F'_{\mu\nu}F^{\mu\nu}$ are assumed negligible for the PT/GW dynamics. The dark and visible sectors are assumed thermalised at a common temperature, $\xi \equiv T_{\text{dark}}/T_{\text{vis}} = 1$, with the dark relativistic degrees of freedom added to g_* .

The **critic-agent** validated the model with *warning* status and *no red flags*. The independent (scanned) parameters are $\{v, g_D\}$; the dependent quantities λ_Φ (CW-fixed by $V'_{\text{eff}}(v) = 0$), $m_{A'} = g_D v$ and $m_\rho = \sqrt{\beta_\lambda} v$ are computed internally by the backend. The model matches the single-field scale-invariant selection rule of **backend/semianalytic_pipeline.py**. The critic raised five warnings and three clarification requests, summarised in Table 1.

Table 1: Critic warnings and clarification requests (source: `handoff_critique.json`).

Item	Note
λ_Φ meaning	Renormalized quartic at $\mu = v$ fixed by $V'_{\text{eff}}(v) = 0$; no implicit tree mass term.
Thermal/Debye masses	Supplied by backend high- T expansion ($\Pi_{A'} \sim (g_D^2/3)T^2$); no daisy resummation or RG running ($\mu = v$).
$\xi = 1$ thermalization	Common-temperature assumption in mild tension with negligible portal/kinetic mixing; affects g_* and N_{eff} .
Higgs portal	Must remain $\lambda_{H\Phi} \ll 10^{-10}$ to preserve conformality.
Gauge dependence	Order-of-magnitude theory uncertainty on $h^2\Omega_{\text{GW}}$ (arXiv:1707.06765); backend uses a fixed gauge.

3 Pipeline status

All pipeline stages produced schema-valid handoffs. Statuses are listed in Table 2.

Table 2: Pipeline stage statuses.

Stage	Status	Note
preliminary-librarian	ok	7 verified papers; model well-covered.
critic	warning	Validated, no red flags; 5 warnings, 3 clarifications.
fopt	ok	Backend executed; 148/157 viable, 8 failed, 1 not_fopt.
pta	ok	Benchmark scan + PTArcade NG15 campaign (10^5 samples).
constraints	warning	No exclusions; 4 high-priority missing calculations.
prior-audit	warning	22 items; high-severity uncontrolled items present.

4 Preliminary literature search

The **preliminary-librarian-agent** identified seven verified papers (Table 3). The exact model (conformal dark $U(1)$ with complex scalar plus dark gauge boson, CW breaking, supercooled FOPT, nHz GW matching PTA) has been studied directly in Gonçalves et al. [1] and Balan et al. [2]; the model structure and target band are therefore *not* novel. The remaining novelty lies in a dedicated, reproducible **fopt-pta** + PTArcade Bayesian campaign for the minimal (DM-free) version with transparent FOPT parameter extraction and gauge-uncertainty bookkeeping.

Benchmark hints adopted downstream: $g_D \in [0.55, 0.9]$ and $v \sim 0.1\text{--}0.7$ GeV (to land in the nHz band), guided by the NANOGrav best-fit $g_D = 0.59$ of [1] and Points A/B of [2].

Table 3: Verified literature (source: `handoff_librarian_preliminary.json`). Class: SM=same model, ST=same target/different model, SS=structurally similar, BG=background.

arXiv	First author	Class	Relevance
2501.11619	Gonalves	SM	Conformal dark $U(1)'$, CW, supercooled FOPT, NANOGrav fit with SMBHB; best-fit $g_D = 0.59$.
2502.19478	Balan	SM	Conformal dark $U(1)'$ + sub-GeV DM fermion; nHz SGWB; Points A/B benchmarks.
2306.17086	Fujikura	ST	nHz SGWB from confining (holographic dilation) conformal PT.
2105.01007	Borah	SS	Dark $U(1)_D$ with explicit (non-conformal) potential, NANOGrav 12.5yr.
1907.08899	Mohamadnejad	SS	Scale-invariant dark $U(1)$ vector DM; mHz LISA band.
2602.14866	Feng	SS	Minimal dark $U(1)$; gauge-independent GW computation.
1707.06765	Chiang	BG	Gauge-dependence of GW from scale-invariant $U(1)'$ FOPT.

5 First-order phase transition results

The `fopt-agent` executed `backend/semianalytic_pipeline.py` (selected by the single-field scale-invariant rule). Inputs: $\chi_0 = v$ [GeV], $\{A' : g_D\}$ boson couplings, $\{A' : 3\}$ boson dof, empty fermion dicts; the CW-fixed quartic and scalar/Goldstone contributions are computed internally. The scan covered $v \in [0.0025, 10]$ GeV and $g_D \in [0.45, 0.9]$. Out of 157 points, 148 are viable, 8 failed (physical or numerical, including eternal-inflation/percolation failures at the lowest g_D) and 1 is not a FOPT; 78 viable points lie in the PTA band. The regime is strongly supercooled with α ranging over many orders of magnitude (up to $\sim 10^{16}$) and $\beta/H \sim 10\text{--}40$. Representative benchmarks are listed in Table 4. The quantity $f_{\text{peak,est}}$ reported by the FOPT backend is a template-independent order-of-magnitude anchor and is *not* the true spectral peak (the latter is owned by the PTA stage).

Table 4: Representative viable FOPT benchmarks at fixed $g_D = 0.5$ (source: `handoff_fopt.json`). T in GeV, f in Hz.

v [GeV]	α	β/H	T_n	T_{reh}	$f_{\text{peak,est}}$
0.05	2.4×10^{14}	15.0	1.8×10^{-6}	4.4×10^{-3}	7.3×10^{-9}
0.10	6.3×10^{14}	14.3	2.9×10^{-6}	8.8×10^{-3}	1.4×10^{-8}
0.173	1.4×10^{15}	13.8	4.2×10^{-6}	1.5×10^{-2}	2.3×10^{-8}
0.30	3.4×10^{15}	13.2	5.9×10^{-6}	2.7×10^{-2}	3.9×10^{-8}
0.50	7.9×10^{15}	12.7	8.1×10^{-6}	4.4×10^{-2}	6.5×10^{-8}
0.692	1.4×10^{16}	12.4	9.8×10^{-6}	6.1×10^{-2}	8.9×10^{-8}

Backend caveats. The backend uses renormalization scale $\mu = v$ with no RG running and no daisy resummation; it employs a high- T (small field/ T) polynomial expansion of the effective potential and a Gaussian percolation approximation. The order-of-magnitude gauge dependence of the CW+thermal effective potential [7] propagates into α , β/H , T_{reh} and hence the GW amplitude/peak.

6 PTA results

The `pta-agent` computed redshifted spectra with `backend/spectrum.py` and scored them against the NG15 free-spectrum violins. Given the strongly supercooled, vacuum-dominated regime ($\alpha \gg 1$, $K = \alpha/(1 + \alpha) \rightarrow 1$, $k_{sw} = 1$), the dissipative bulk-flow (`dbf`) template of Lewicki & Vaskonen [8] was selected; the higgsless sound-wave template [9] is outside its validity domain for $\alpha \gg 1$ and was retained only for comparison. A scan of 1271 viable points placed the best 14/14-bin window interior at $v \in [0.35, 2.4]$ GeV, $g_D \in [0.51, 0.63]$. The four top-scoring (14/14) benchmark points are listed in Table 5.

Table 5: Top-scoring (14/14) benchmark spectra (source: `handoff_pta.json`). f_{peak} in Hz.

Rank	v [GeV]	g_D	α	β/H	T_* [GeV]	f_{peak}
1	0.748	0.560	1.2×10^9	21.4	0.064	1.9×10^{-8}
2	2.386	0.510	3.5×10^{15}	12.4	0.215	3.0×10^{-8}
3	0.508	0.630	8.3×10^4	35.8	0.048	2.3×10^{-8}
4	0.345	0.540	6.1×10^{10}	19.0	0.033	7.9×10^{-9}

PTArcade NG15 campaign. A PTArcade campaign was run in `ceffyl` mode on the NG15 dataset with priors $\log_{10} v \sim \mathcal{U}(-3, 1.699)$ [GeV] and $g_D \sim \mathcal{U}(0.45, 0.90)$, fixing the `dbf` template, $\xi = 1$, $k_{sw} = 1$ and $n_{A'} = 3$. The run used $N_{\text{samples}} = 10^5$ at acceptance 0.385 (~ 2018 s); the posterior is interior to the priors with no boundary contact at 3σ . The maximum-a-posteriori (MAP) point is $v = 1.09$ GeV ($\log_{10} v = 0.038$), $g_D = 0.528$. The Bayesian-estimate means are $v = 0.871$ GeV, $g_D = 0.527$. Credible regions are given in Table 6; the 68% region maps to a GeV-scale dark photon $m_{A'} \in [0.31, 0.67]$ GeV and a light scalon $m_\rho \in [10.7, 24.4]$ MeV.

Table 6: PTArcade NG15 credible regions (source: `handoff_pta.json`).

Region	$\log_{10} v$ low	$\log_{10} v$ high	g_D low	g_D high
1 σ (68%)	-0.214	0.091	0.508	0.546
2 σ (95%)	-0.476	0.165	0.504	0.587
3 σ (99.7%)	-0.820	0.215	0.503	0.636

7 Constraints

The `constraint-agent` evaluated 12 constraints on the PTArcade 1 σ region ($g_D \in [0.508, 0.546]$, $v \in [0.61, 1.23]$ GeV; $m_{A'} \in [0.31, 0.67]$ GeV, $m_\rho \in [10.7, 24.4]$ MeV, $T_{\text{reh}} \sim 60\text{--}100$ MeV). **No constraint excludes any part of the region.** Of the 12 constraints, 1 is directly testable, 2 approximately, 3 require recasting, 4 require new backend calculations, and 2 are not applicable. A key physics point is that $2m_\rho \sim 21\text{--}49$ MeV $\ll m_{A'} \sim 0.3\text{--}0.7$ GeV, so the dark decay $A' \rightarrow \rho\rho$ is kinematically open and can render A' largely invisible to visible-decay searches. SN1987A (C07) is *not applicable* because the preferred dark-photon mass lies above its $\lesssim 100$ MeV reach. The central internal tension flagged by the critic and constraint agents is that $\xi = 1$ (which feeds g_* and the GW redshift) requires a nonzero portal, whereas both ϵ and $\lambda_{H\Phi}$ are set $\simeq 0$ for the GW dynamics; this gates most cosmological/laboratory bounds. The constraints are summarised in Table 7.

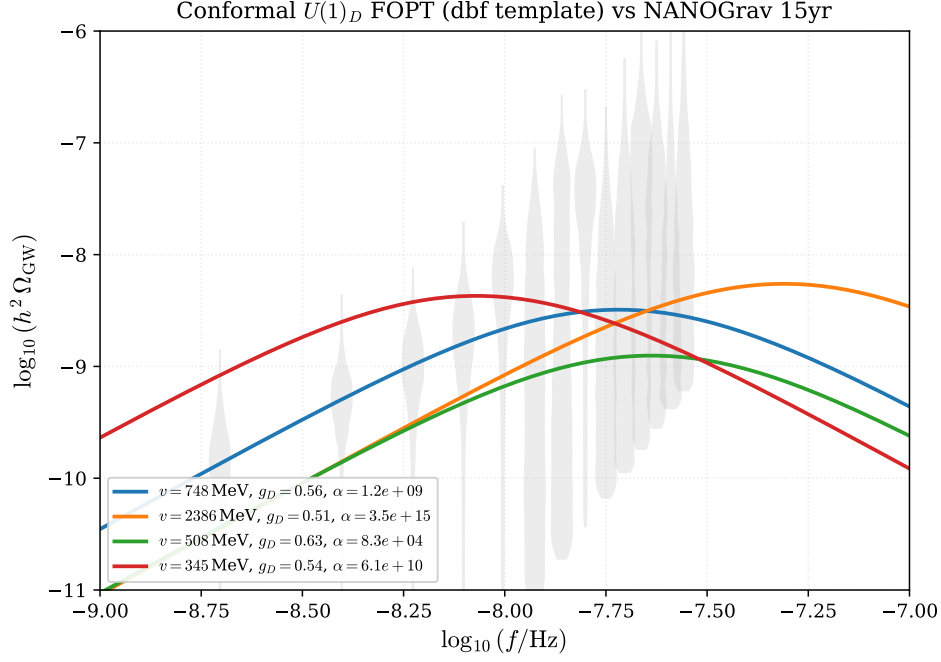


Figure 1: Estimated benchmark $h^2\Omega_{\text{GW}}(f)$ overlaid on the NG15 PTA violins (source: `pta_benchmark_spectrum.pdf`).

Table 7: Constraints on the 1σ region (source: `handoff_constraints.json`). Class: D=direct, A=approximate, R=recast, N=new backend, NA=not applicable.

ID	Constraint	Class	Verdict	Source
C01	BBN MeV-reheating ($T_{\text{reh}} > 3 \text{ MeV}$)	D	allowed ($T_{\text{reh}} \sim 60\text{--}100 \text{ MeV}$)	2502.19478
C02	ΔN_{eff} (dark radiation)	N	pending	2502.19478
C03	Thermalization / $\xi = 1$ justification	N	pending (ϵ_{min})	2502.19478
C04	Visible-decay collider (A')	R	pending (ϵ , BR)	2603.08430
C05	Electron beam-dump (E137 etc.)	R	pending (ϵ , τ)	1406.2698
C06	Invisible / missing-energy (A')	R	pending (ϵ , BR)	2603.08430
C07	SN1987A energy loss	NA	not applicable ($m_{A'} > 0.1 \text{ GeV}$)	1611.03864
C08	Scalon late decay (BBN/CMB)	N	pending (τ_ρ , Y_ρ)	2502.19478
C09	Scalon relic overabundance	N	pending ($\langle\sigma v\rangle$)	2502.19478
C10	Higgs portal / $h \rightarrow \text{inv}$	A	allowed ($\lambda_{H\Phi} \ll 10^{-10}$)	2501.11619
C11	Perturbativity of g_D	D	allowed ($\alpha_D \sim 0.02$)	1707.06765
C12	Dark-photon direct detection	NA	not applicable (A' not DM)	2603.08430

Missing backend calculations. Four high-priority calculations gate constraints C02–C09: (M1) minimum kinetic mixing ϵ for $\xi = 1$ thermalization; (M2) dark-photon decay widths/lifetime and $A' \rightarrow \rho\rho$ vs visible branching ratios; (M3) ΔN_{eff} for the thermalized MeV-scale dark sector; (M4) scalon relic yield, lifetime and BBN/CMB energy injection.

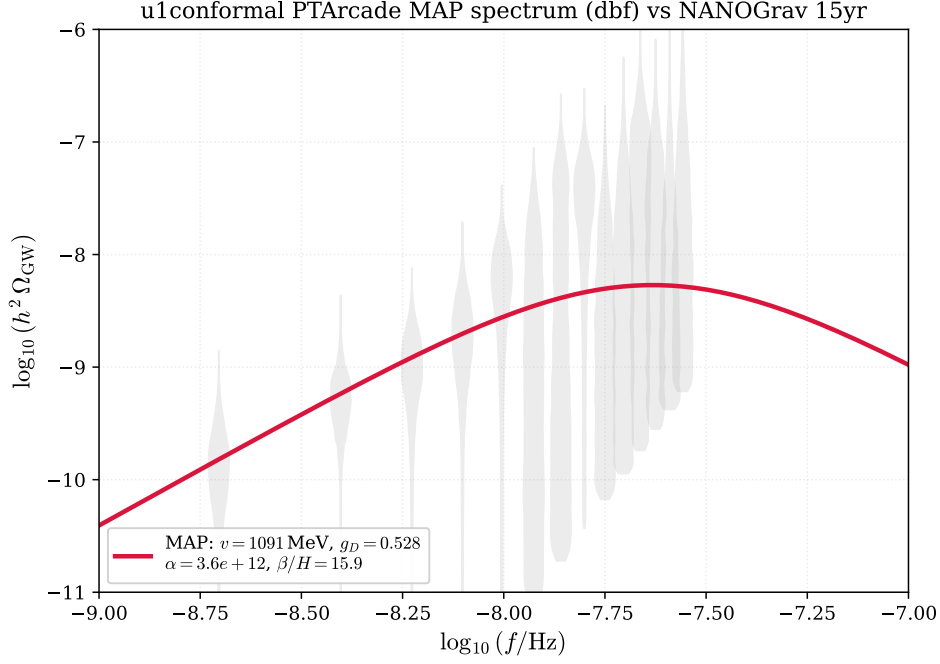


Figure 2: Spectrum at the PTArcade Bayesian MAP point ($v = 1.09$ GeV, $g_D = 0.528$) versus frequency with the NG15 violins (source: `pta_ptarcade_spectrum.pdf`).

8 Assumptions and limitations

The **prior-agent** audited 22 items (severity distribution peaks at 5; four items at severity 8 and one at severity 9). The highest-severity and uncontrolled items are collected in Table 8. The three dominant limiting factors are: (i) the gauge-dependence theory uncertainty (A07, order-of-magnitude on $h^2\Omega_{\text{GW}}$); (ii) the $\xi = 1$ versus $\epsilon \simeq 0$ internal tension (A05) and the associated cosmology calculations (A19, A20, A21); and (iii) the no-SMBHB-foreground PTArcade fit (A17), since NANOGrav new-physics preferred regions depend strongly on astrophysical-foreground modelling. Consequently **the preferred region is provisional**.

9 Conclusions

The classically scale-invariant dark $U(1)_D$ model with CW radiative breaking produces a strongly supercooled FOPT whose SGWB falls in the nHz band probed by PTAs. A PTArcade NG15 campaign with the physically motivated **dbf** template prefers $v \simeq 1.09$ GeV, $g_D \simeq 0.528$ (MAP), i.e. a GeV-scale dark photon and an MeV-scale scalon, in line with the literature scale for this model class [1, 2].

Robust: the model is theoretically consistent (no anomalies, correct dof bookkeeping, perturbative g_D , C11); the FOPT is strongly supercooled and lands in the nHz band; the **dbf** template is the correct strong-supercooling choice; the BBN reheating floor (C01) and Higgs-portal/perturbativity bounds (C10, C11) are satisfied; SN1987A and dark-photon direct detection are not applicable (C07, C12).

Conditional / provisional: the preferred region depends on the no-SMBHB fit (A17) and carries an order-of-magnitude gauge-dependence uncertainty (A07); no constraint currently excludes it, but most cosmological/laboratory bounds are not yet applicable.

Blocked / uncontrolled: the $\xi = 1$ versus $\epsilon \simeq 0$ tension (A05) and the four missing calculations (A19; ϵ_{min} , A' branchings, ΔN_{eff} , scalon relic/decay) must be resolved before the viability of the region can be confirmed. Recommended follow-ups include refitting with an SMBHB foreground, propagating a gauge-variation band, and running the four dark-sector cosmology

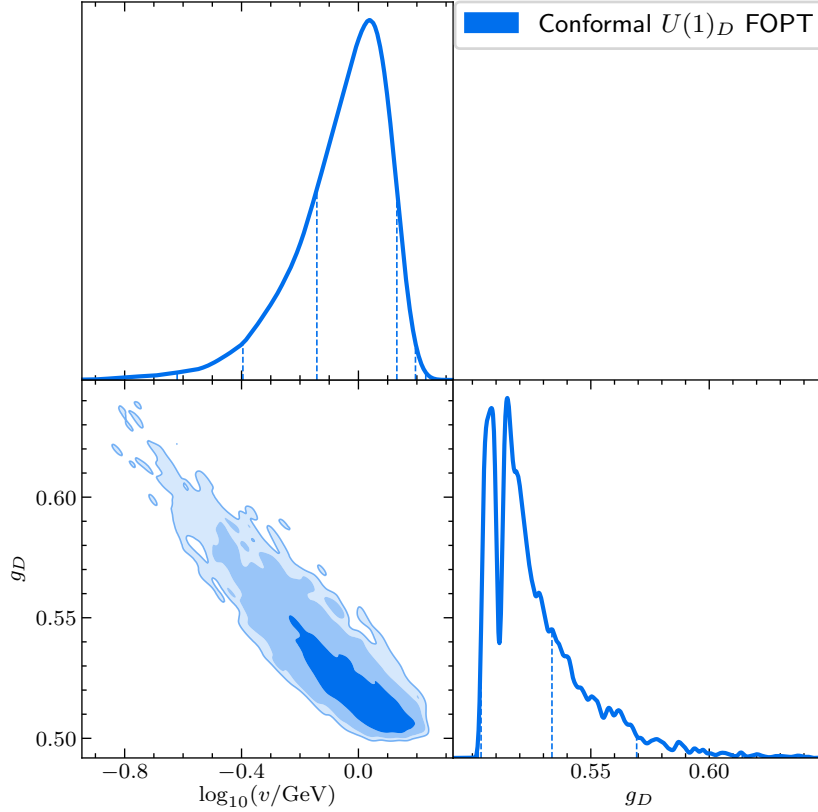


Figure 3: Bayesian posterior distributions of $(\log_{10} v, g_D)$ from the PTArcade NG15 `ceffyl` campaign (source: `u1conformal-posteriors.pdf`).

modules.

10 Inconsistencies

No contradictory numerical claims were found across handoffs. One physics-level internal tension is propagated consistently by all agents and is *not* silently resolved here: $\xi = 1$ (used by the FOPT/PTA stages to set g_* and the GW redshift) is incompatible with exactly vanishing portals ($\epsilon = \lambda_{H\Phi} = 0$); a minimum nonzero ϵ is required and remains to be computed (A05, C03).

References

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Table 8: High-severity (≥ 8) and selected assumptions/limitations (source: `handoff_prior_audit.json`). Ctrl=controlled.

ID	Sev.	Ctrl	Description
A19	9	no	Four high-priority missing calculations gate C02–C09.
A05	8	no	$\xi = 1$ requires a portal, but $\epsilon, \lambda_{H\Phi} \simeq 0$ (central internal tension).
A07	8	no	Gauge dependence of CW+thermal V_{eff} : order-of-magnitude on $h^2\Omega_{\text{GW}}$ (1707.06765).
A17	8	no	Model-only (no SMBHB foreground) fit to NG15; preferred region provisional.
A20	8	no	Scalon (11–24 MeV) fate: overclosure (if stable) or late decay (if $\epsilon \neq 0$).
A03	7	no	Kinetic mixing $\epsilon \simeq 0$ for PT/GW; nonzero on cosmology side.
A11	7	yes	Very large α ($\sim 10^{16}$); eternal-inflation/ percolation failures excluded.
A21	7	no	ΔN_{eff} likely strongest test of $\xi = 1$.
A08	6	no	No RG running / no daisy resummation ($\mu = v$).
A12	6	yes	dbf template choice; higgsless out of domain.
A15	4	yes	PTArcade priors; posterior interior, no boundary contact.

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